Harsh Gupta

Robotics & Perception Engineer | ROS2 | Computer Vision | Edge Deployment | Autonomous Systems

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Education

IIITDM Kancheepuram 2022 – 2026
Bachelor of Technology in Mechanical Engineering Chennai, Tamil Nadu

Professional Experience

Software Developer Intern June 2025 – December 2025
Unico International Pvt. Ltd., Bangalore

- Designed an **LLM-powered Task Management System** with event-driven workflow orchestration, automated task assignment, and real-time synchronization across distributed services.
- Built a scalable **Forex Trading Application** with low-latency market data ingestion, trade execution pipelines, order management, and P&L tracking using FastAPI and Supabase.

Technical Projects

Visual Servoing for Mobile Robot Goal Seeking | Github Link

- Built a **ROS2 perception and navigation stack** on **TurtleBot3** in **Gazebo**: **OpenCV** HSV masking, contour detection, and centroid-based error estimation at **30 Hz** from live camera frames.
- Two-node **perception pipeline** — detector publishes tracking error; navigator applies a **P-controller** on **cmd_vel** for alignment and approaching; added **out-of-frame recovery** via last-seen direction bias and timed search.

Driver Drowsiness Detection | Github Link

- Built a **real-time drowsiness detection system** using **YOLOv8**, **OpenCV**, and **MediaPipe** achieving **25+ FPS**; trained dual models on **53K+ images** with **GroundingDINO** auto-labeling, cutting annotation effort by 60%.
- Integrated inference with a **PyQt5 GUI** and audio alerts; optimized for **edge deployment** via **ONNX** and **TensorRT**; cross-validated with **TensorFlow/Keras** and PyTorch workflows.

Brain Tumor Segmentation using U-Net | Github Link

- Developed a **U-Net semantic segmentation model** for brain tumor detection on the Kaggle LGG dataset (110 patients, 3929 MRI images); applied augmentation and normalization for robust generalization.
- Achieved **Dice: 0.83** and **IoU: 0.76**; enhanced model interpretability via **Grad-CAM** and deployed a real-time demo using **Streamlit**.

Position of Responsibility

Member, CS Team July 2023 – December 2024
Astra – Drone Racing Club, IIITDM Kancheepuram

- Trained a custom **YOLO-based object detection model** on competition-specific datasets for autonomous UAV tasks; deployed on **NVIDIA Jetson Nano** at **20+ FPS** for onboard inference without ground-side processing.
- Integrated **perception pipeline** in **ROS2**; used **MAVROS** for MAVLink-based offboard control enabling autonomous target approach; validated workflows in **Gazebo** before hardware deployment.

Technical Skills

- **Computer Vision and Robotics:** OpenCV, YOLOv8, Object Detection, Multi-Object Tracking (DeepSORT), Semantic Segmentation, CNNs, MediaPipe, GroundingDINO, GStreamer, ROS2, Gazebo
- **Deep Learning & Optimization:** PyTorch, TensorFlow/Keras, U-Net, Grad-CAM, ONNX, TensorRT, Edge Deployment, Model Quantization
- **Programming & Tools:** Python, C++, C, JavaScript, Git, Linux, FastAPI, Supabase, Jupyter Notebook, Docker
- **Coursework:** Computer Vision, Data Science, Modern C++ (OOPs)